Seminar in
Laboratory of Visual Intelligence and Pattern Analysis (VIPA)

# A Review on Normalization Methods

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# Outline

- Introduction
- Taxonomy
- Normalization Techniques: Methods and Analysis
- Applying Normalization Methods in Practice
- Challenges and Future Work

# Introduction: Why this topic?

- **Tasks** →

  Super Resolution  Knowledge Transfer  …
  Neural Architecture Search  Lifelong Learning

- **Models** →

  Capsule Network  Densely Connected Network  …
  Squeeze-and-Excitation Network

Previously Covered Topics

# Introduction: Why this topic?

- **Tasks** →
  - Super Resolution
  - Knowledge Transfer
  - Neural Architecture Search
  - Lifelong Learning
  - …
- **Models** →
  - Capsule Network
  - Densely Connected Network
  - Squeeze-and-Excitation Network
  - …

Previously Covered Topics

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- **Modules** →
  - Normalization Method

# Introduction: What?

- **Normalization**
  - A tool used in science and engineering for decades
  - **Standardizing** different statistical properties of a representation

$$\hat{\boldsymbol{x}} = \frac{\boldsymbol{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}$$

# Introduction: What?

- **Normalization**
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$$\hat{x} = \frac{x - \mu}{\sigma}$$

- **Normalization in Deep Learning**

  **=** Feature Normalization **+** Weight Normalization **+** Gradient Normalization

  - **Goals:**
    - Avoid **vanishing** and **exploding** gradients
    - Remove **redundant** and **noisy** features
    - Improve **generalization performance** and **convergence speed**

# Taxonomy of Normalization Methods

## Normalization Methods in Deep Learning

**Feature Normalization**

Batch Normalization
Instance Normalization
Layer Normalization
Group Normalization
Divisive Normalization
Switchable Normalization

**Weight Normalization**

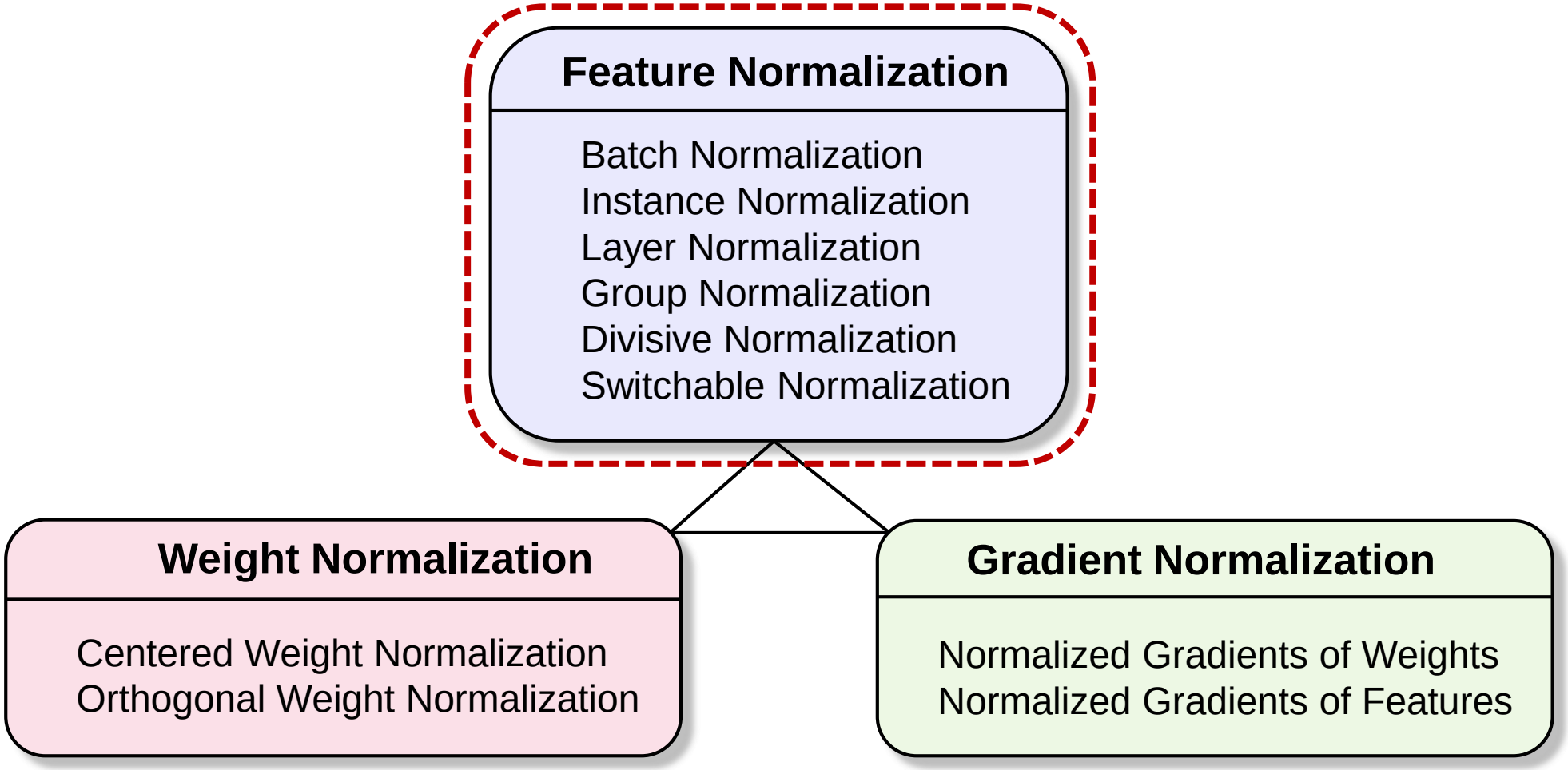
Centered Weight Normalization
Orthogonal Weight Normalization

**Gradient Normalization**

Normalized Gradients of Weights
Normalized Gradients of Features

# Taxonomy of Normalization Methods

## Normalization Methods in Deep Learning



# Reading List

- **Feature Normalization**
  - **Batch Normalization**: Accelerating deep network training by reducing internal covariate shift, *ICML 2015*
  - **Batch Renormalization**: Towards reducing minibatch dependence in batch-normalized models, *NIPS 2017*
  - **Adaptive Batch Normalization**: Revisiting batch normalization for practical domain adaptation, *PR 2018*
  - **Instance Normalization**: The missing ingredient for fast stylization, *CVPR 2018*
  - **Conditioned Instance Normalization:** A Learned Representation For Artistic Style, *ICLR 2017*
  - **Adaptive Instance Normalization:** Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization, *ICCV 2017*
  - **Layer Normalization**, *2016* **(by Hinton)**
  - **Group Normalization**, *ECCV 2018* **(Best Paper Honorable Mention)**
  - **Divisive Normalization:** Normalizing the normalizers-comparing and extending network normalization schemes, *ICLR 2017*
  - **Switchable Normalization**: Differentiable Learning to Normalize via Switchable Normalization, *2018*

# Reading List

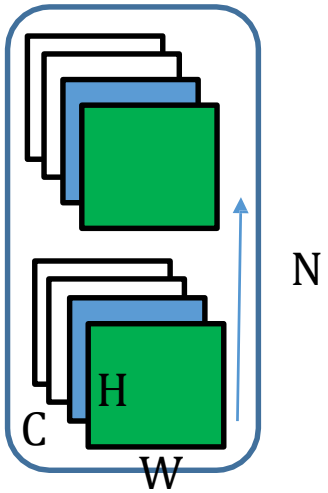
- **Weight Normalization**
  - Normalization Propagation: A Parametric Technique for Removing Internal Covariate Shift in Deep Networks. *ICML 2016*.
  - Exploring Normalization in Deep Residual Networks. *AAAI 2017*
  - Weight Normalization: A Simple Reparameterization to Accelerate Training of Deep Neural Networks. *NIPS 2016*
  - Centered Weight Normalization in Accelerating Training of Deep Neural Networks. *ICCV 2017*.
  - Orthogonal Weight Normalization: Solution to Optimization over Multiple Dependent Stiefel Manifolds in Deep Neural Networks. *AAAI 2018*
- **Gradient Normalization**
  - EigenNet: Towards Fast and Structural Learning of Deep Neural Networks. *IJCAI 2018*
  - GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep Multitask Networks. *ICML 2018*

# Feature Normalization

- **Overview**

$$\hat{x} = \frac{x - \mu}{\sigma} \qquad x \in \mathcal{R}^{N \times C \times H \times W}$$

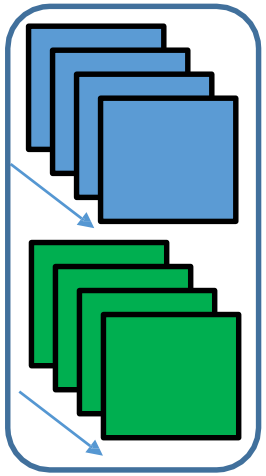
① Batch Norm




$\mu_c = mean(N, H, W)$

$\sigma_c = std(N, H, W)$

② Layer Norm



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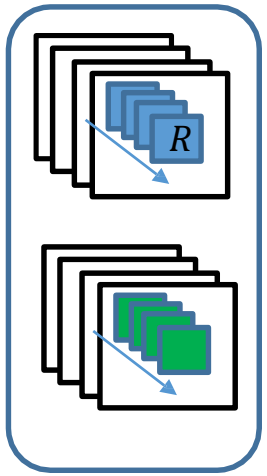
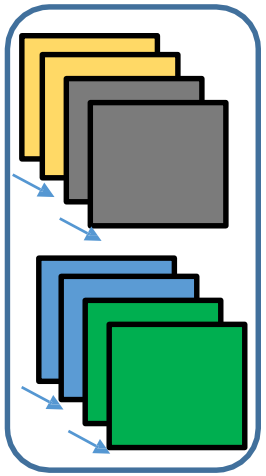
③ Divisive Norm




$\mu_{nR} = mean(R_{C,H,W})$

$\sigma_{nR} = std(R_{C,H,W})$

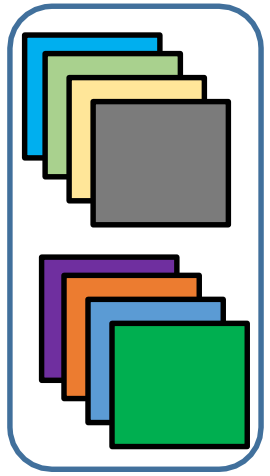
④ Group Norm



$\mu_{nC_i} = mean(C_i, H, W)$

$\sigma_{nC_i} = std(C_i, H, W)$

⑤ Instance Norm

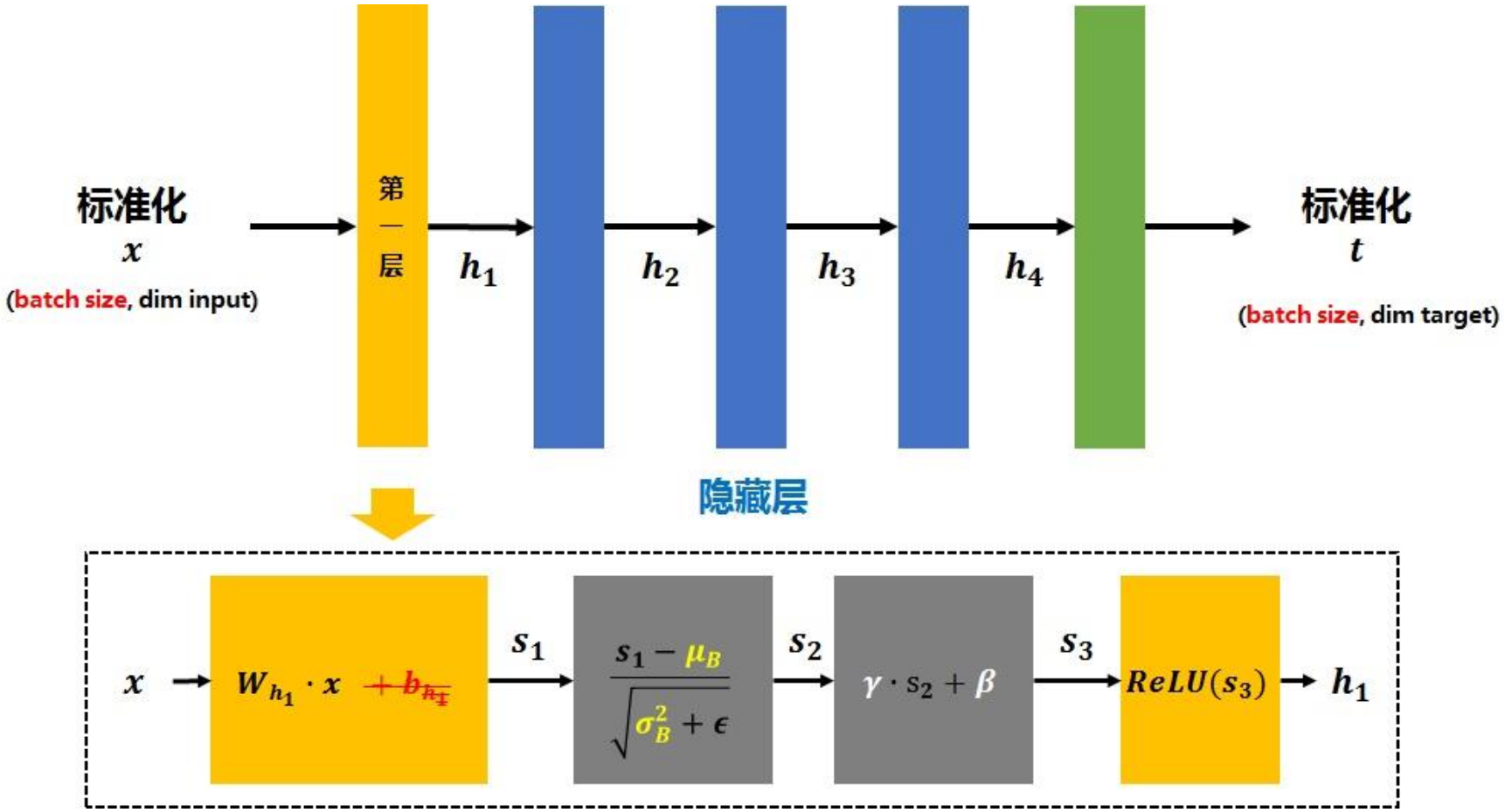


$\mu_{nc} = mean(H, W)$

$\sigma_{nc} = std(H, W)$

# Feature Normalization

- **Batch Normalization: Training Phase**



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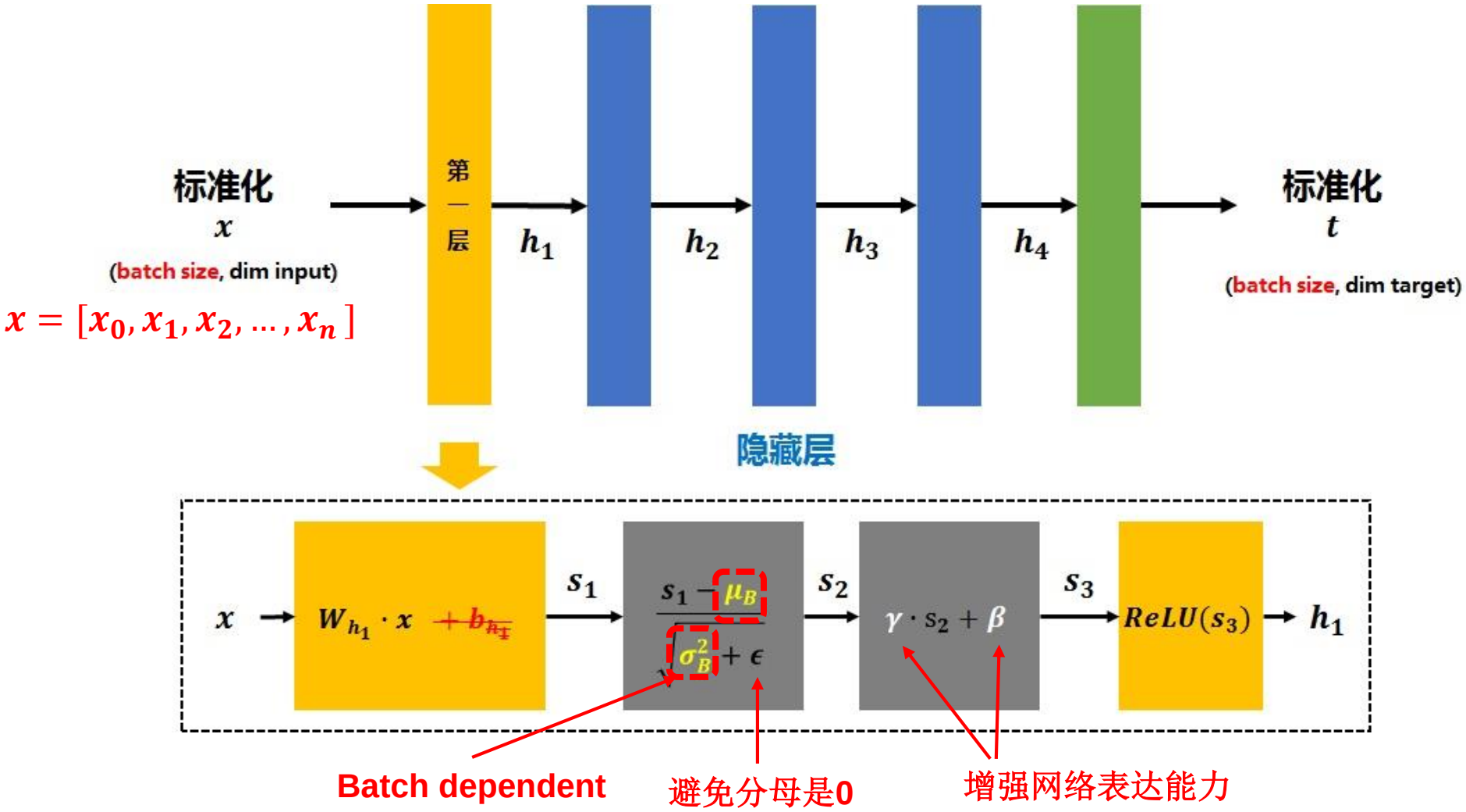
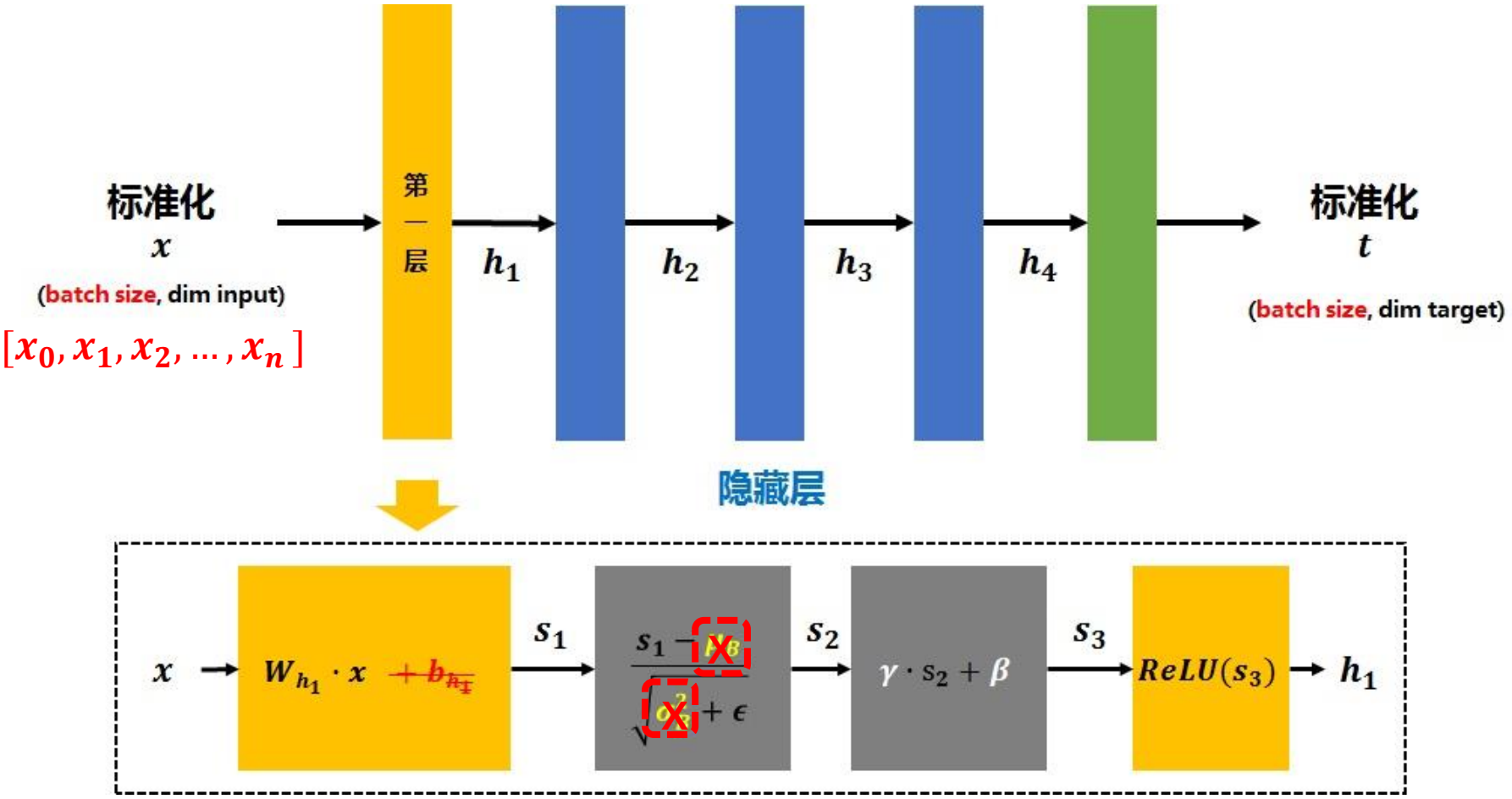


# Feature Normalization

- **Batch Normalization: Inference Phase**




**Inference Phase:** $\mu_{training_data}$ $\sigma_{training_data}$

在训练和测试的时候，实际上是用了不同的值在进行normalization

# Feature Normalization

- **Batch Normalization (BN)**
  - **Benefits:**
    - Make **optimization landscape** significantly more **smooth**[1]
    - Prevent **vanishing** and **exploding** gradients
    - Robustness to different settings of hyperparameters
    - …

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  - **Misunderstanding:**

    ✘ BN reduces internal covariate shift[1,2]. → Title of BN Paper: Accelerating deep network training by reducing internal covariate shift, *ICML 2015*

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    - ✘ BN prevents overfitting[3].


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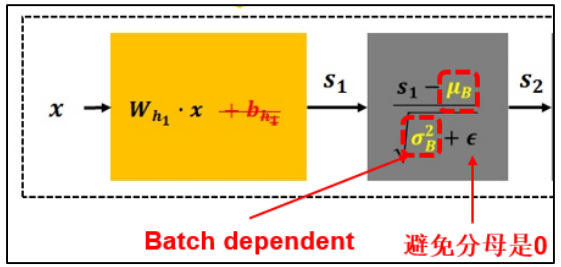
2. https://www.zhihu.com/question/38102762 by 魏秀参
3. Zhang et al. Understanding deep learning requires rethinking generalization, ICLR 2017 (**Best Paper**)

# Feature Normalization

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  - Require a sufficiently **large batch size** to avoid inaccurate estimate of the **batch statistics**.

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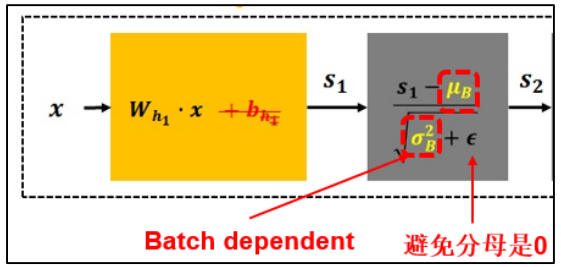


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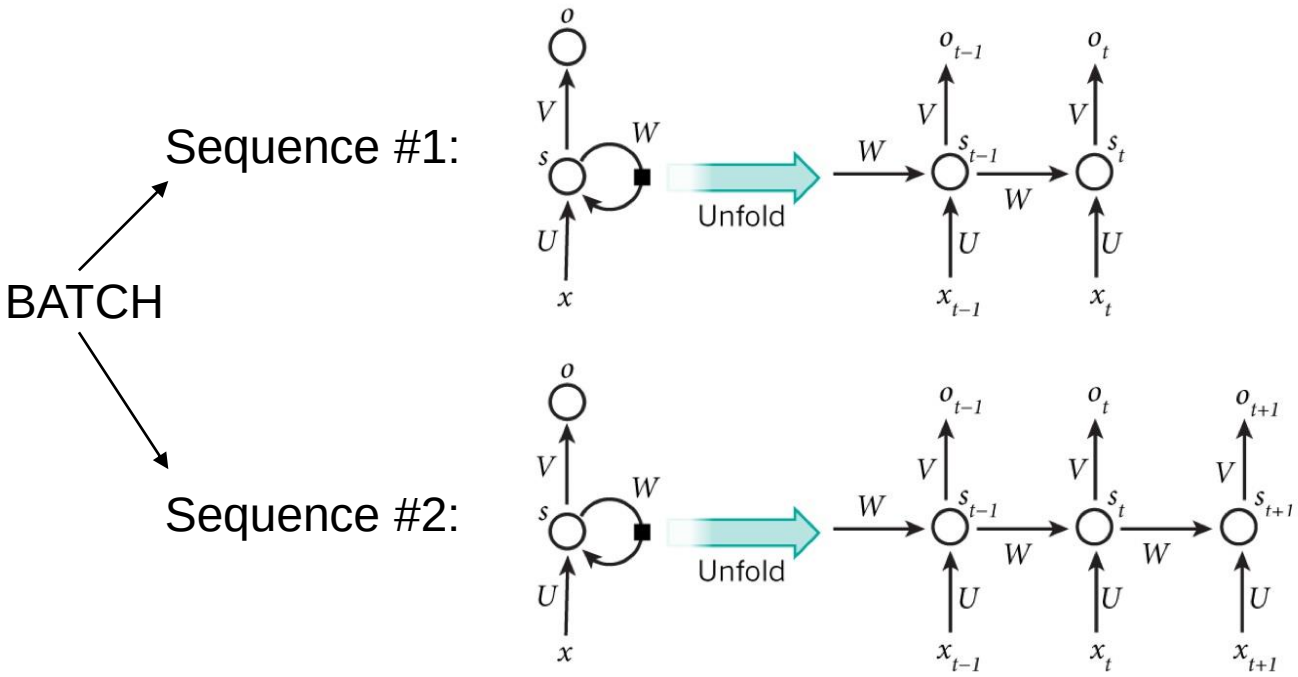
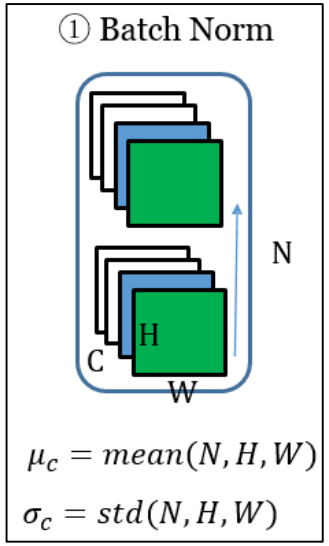
    - Because the inputs to RNN often **vary** with the length of the sequence.



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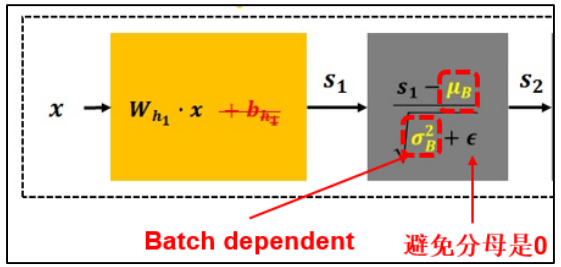


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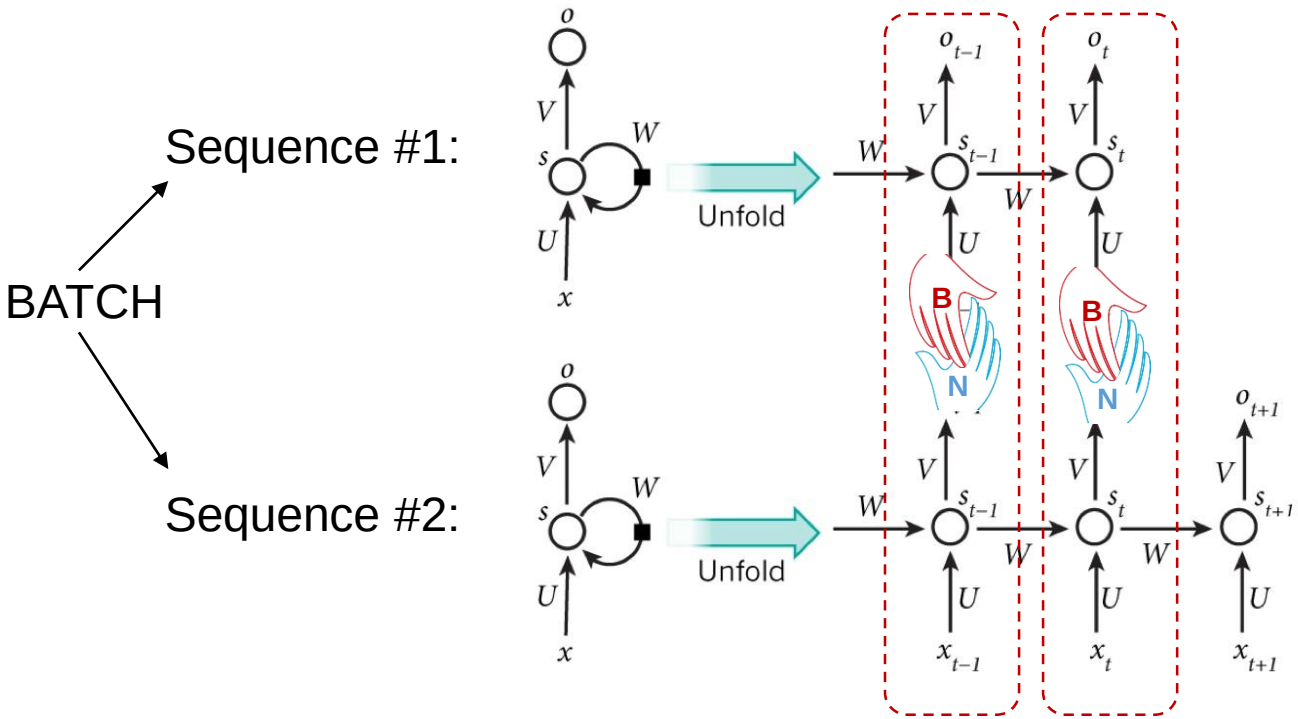


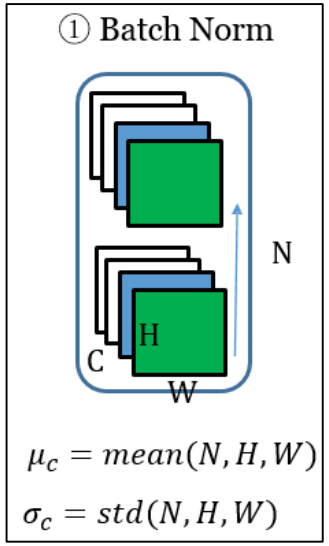

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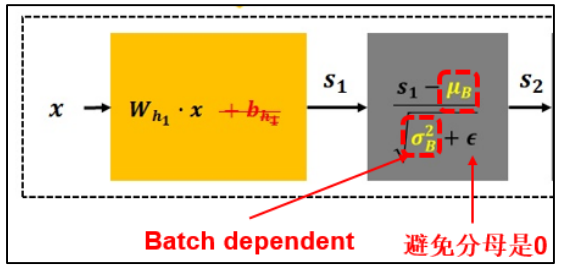


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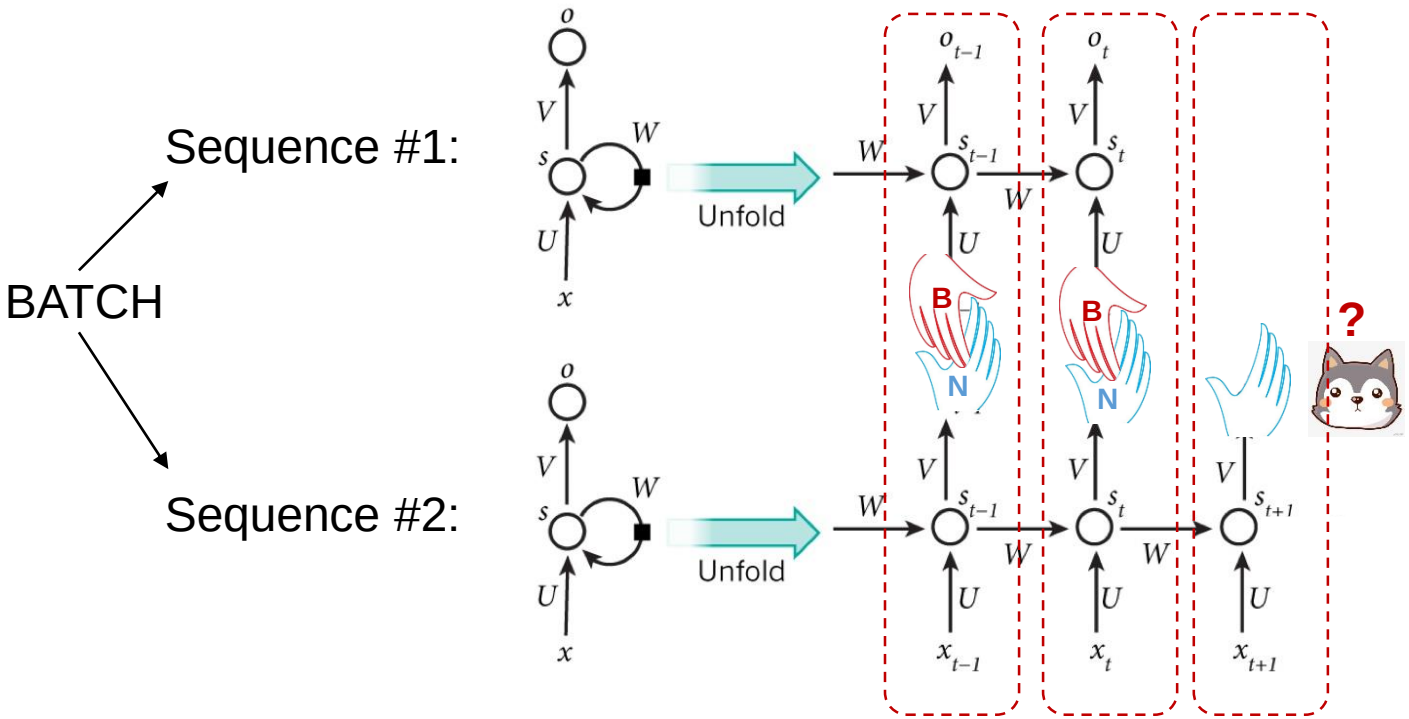
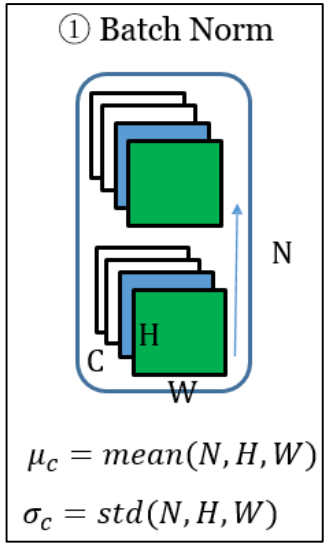
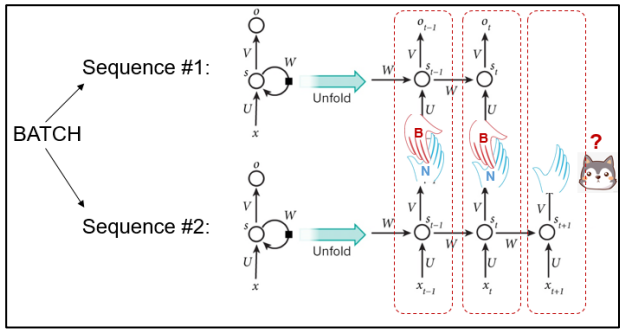


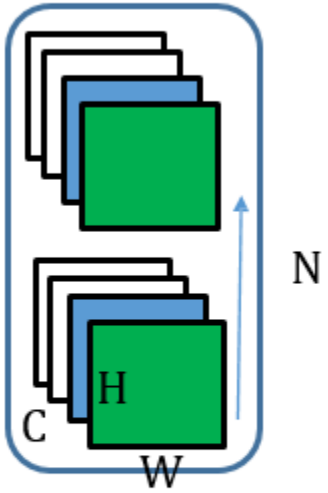

1. Laurent et al. Batch Normalized Recurrent Neural Networks, ICASSP 2016

# Feature Normalization




- **Layer Normalization (LN)**
  - Compute the layer normalization statistics along the $(C, H, W)$ axes for **each sample**.
  - The computation of LN is **independent** of **batch sizes**.

① Batch Norm




$$\mu_c = mean(N, H, W)$$

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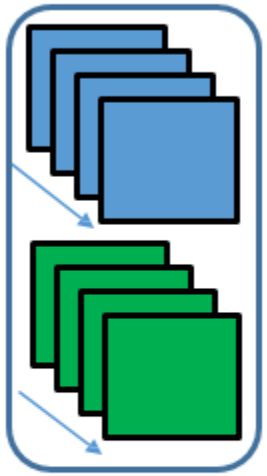

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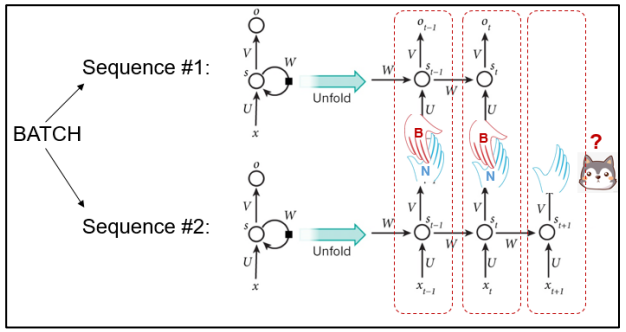
# Feature Normalization

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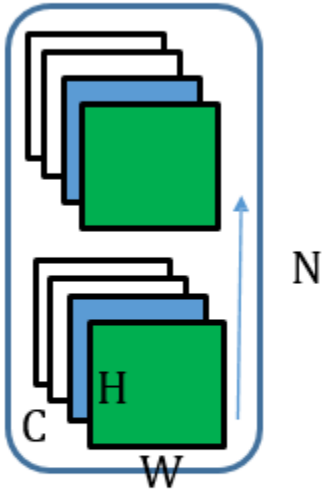


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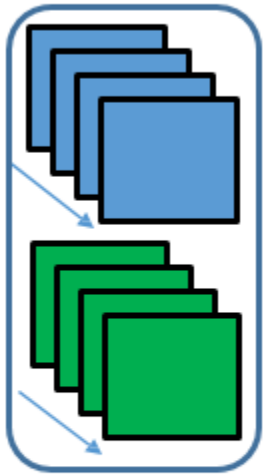


$$\mu_n = mean(C, H, W)$$

$$\sigma_n = std(C, H, W)$$

**Performance**:
- ✓ Works **well** for **RNN**.
- ✓ For **CNN**, direct application of LN leads to a **degradation** of performance.

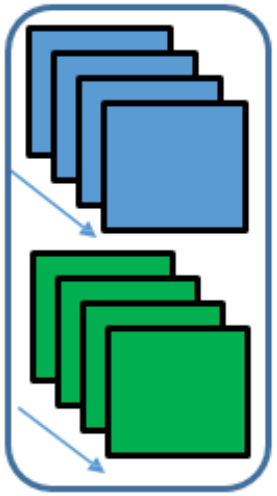
The authors hypothesize that since the statistics in convolutional layers can **vary** quite a bit spatially, normalization with statistics from **an entire layer** might be suboptimal.

# Feature Normalization

- **Divisive Normalization (DN)**
  - A new local normalization schemes, operating on a **local spatial window** across filter channels.

$5 \times 5, 3 \times 3$

② Layer Norm



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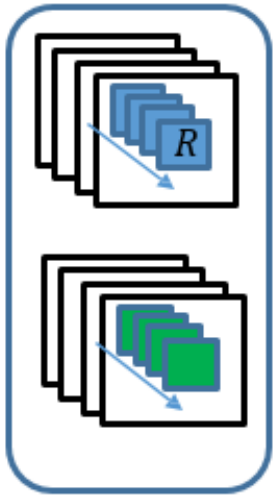
③ Divisive Norm




$$\mu_{nR} = mean(R_{C,H,W})$$

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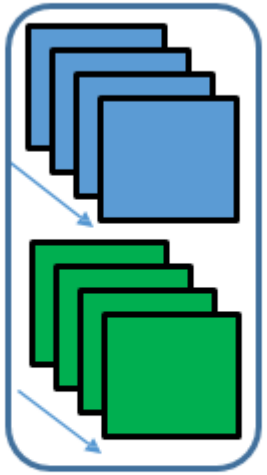
**Performance**:

✓ DN achieves **comparable or slightly better** performance on image classification, RNN language modelling, and image super-resolution, compared with BN and LN.

# Feature Normalization

- **Group Normalization (GN)** (Best Paper Honorable Mention, ECCV 2018)
  - **Group-wise normalization scheme**, inspired by the fact that many classical features like SIFT and HOG are group-wise features.
  - **GN layer** divides channels into **groups** and normalizes the features within each group.

② Layer Norm



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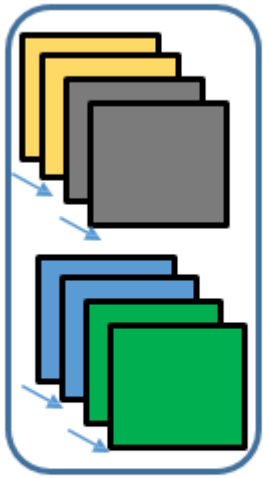

$$\mu_{nC_i} = mean(C_i, H, W)$$

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Group # is 32 by default.

# Feature Normalization

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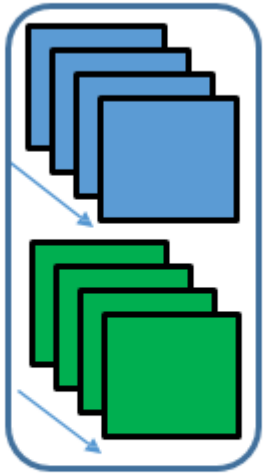

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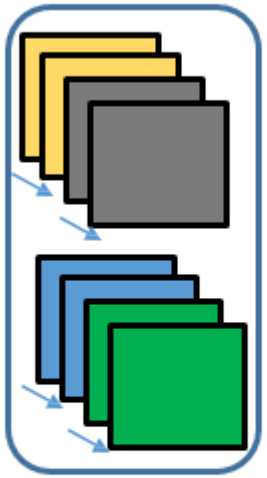

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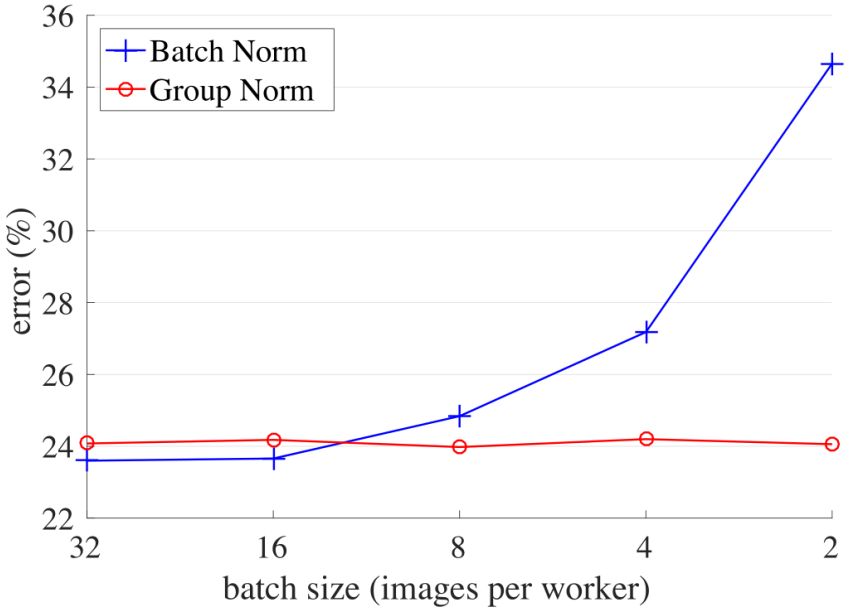
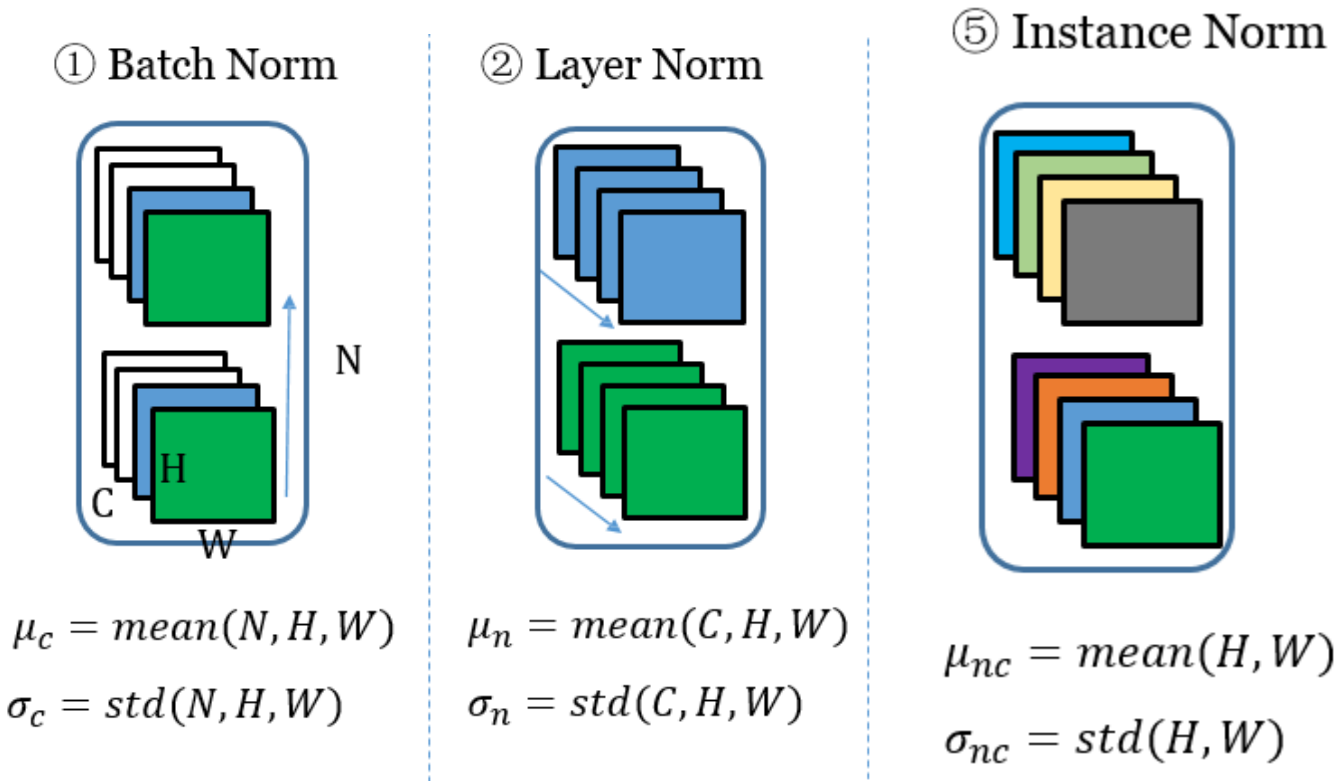



ImageNet classification performance

# Feature Normalization

- **Instance Normalization (IN)**
  - Instance-specific normalization.




**Performance**:
- ✓ Works well in **generative models**.
- ✓ May work well in some other tasks such as Crowd Counting[1].

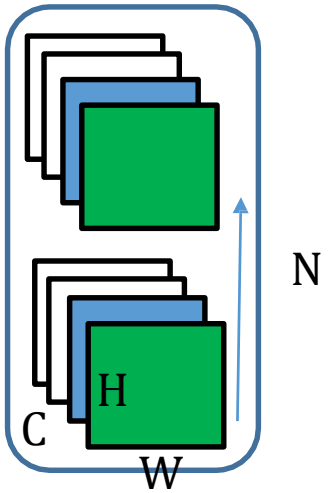
1. Cao et al. Scale Aggregation Network for Accurate and Efficient Crowd Counting, ECCV 2018

# Feature Normalization

- **Overview**

$$\hat{x} = \frac{x - \mu}{\sigma} \qquad x \in \mathcal{R}^{N \times C \times H \times W}$$

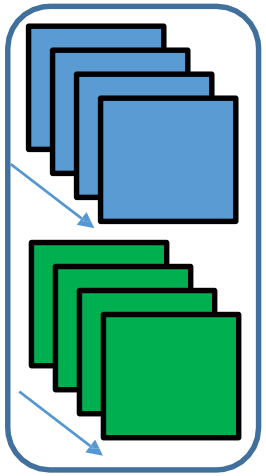
① Batch Norm




$\mu_c = mean(N, H, W)$

$\sigma_c = std(N, H, W)$

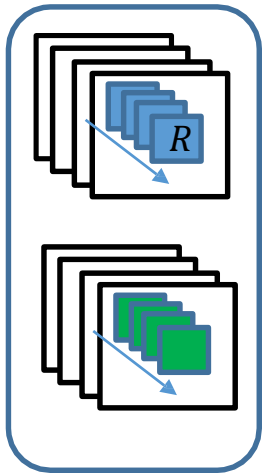
② Layer Norm



$\mu_n = mean(C, H, W)$

$\sigma_n = std(C, H, W)$

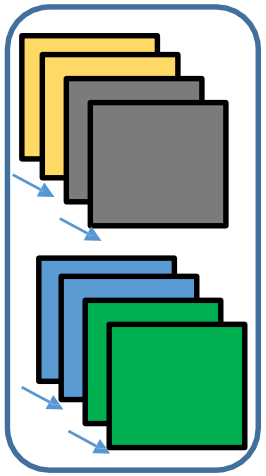
③ Divisive Norm




$\mu_{nR} = mean(R_{C,H,W})$

$\sigma_{nR} = std(R_{C,H,W})$

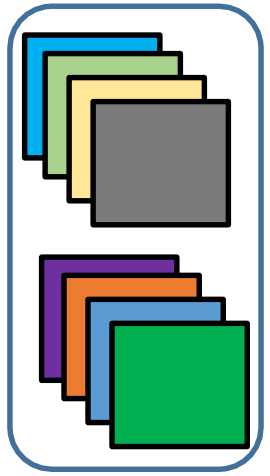
④ Group Norm



$\mu_{nC_i} = mean(C_i, H, W)$

$\sigma_{nC_i} = std(C_i, H, W)$

⑤ Instance Norm



$\mu_{nc} = mean(H, W)$

$\sigma_{nc} = std(H, W)$

# Feature Normalization

- **Variants of BN**
  - **Batch Renormalization** by the same author of BN
    - ✓ 假设每个batch的均值和方差来代替整体训练集的均值和方差之后，可以再通过一个**线性变换**来逼近数据的真实分布：

$$\frac{x_i - \mu}{\sigma} = \frac{x_i - \mu_{\mathcal{B}}}{\sigma_{\mathcal{B}}} \cdot r + d.$$

  - **Adaptive Batch Normalization** (AdaBN) by 王乃岩
    - ✓ Designed for **domain adaption**.
    - ✓ Estimate the mean and variance statistics in BN on a **per-domain basis**, i.e., training phase: using source domain BN statistics; testing phase, using target domain BN statistics.

# Feature Normalization

- **Variants of IN**
  - **Conditioned Instance Normalization** (CIN) by google brain

    **Adaptive Instance Normalization** (AdaIN) by Serge Belongie

    ✓ Both are designed for Neural Style Transfer.

$$\mathrm{CIN}(\mathcal{F}(I_c), s) = \gamma^s \left( \frac{\mathcal{F}(I_c) - \mu(\mathcal{F}(I_c))}{\sigma(\mathcal{F}(I_c))} \right) + \beta^s$$

$$\mathrm{AdaIN}(\mathcal{F}(I_c), \mathcal{F}(I_s)) = \sigma(\mathcal{F}(I_s)) \left( \frac{\mathcal{F}(I_c) - \mu(\mathcal{F}(I_c))}{\sigma(\mathcal{F}(I_c))} \right) + \mu(\mathcal{F}(I_s))$$

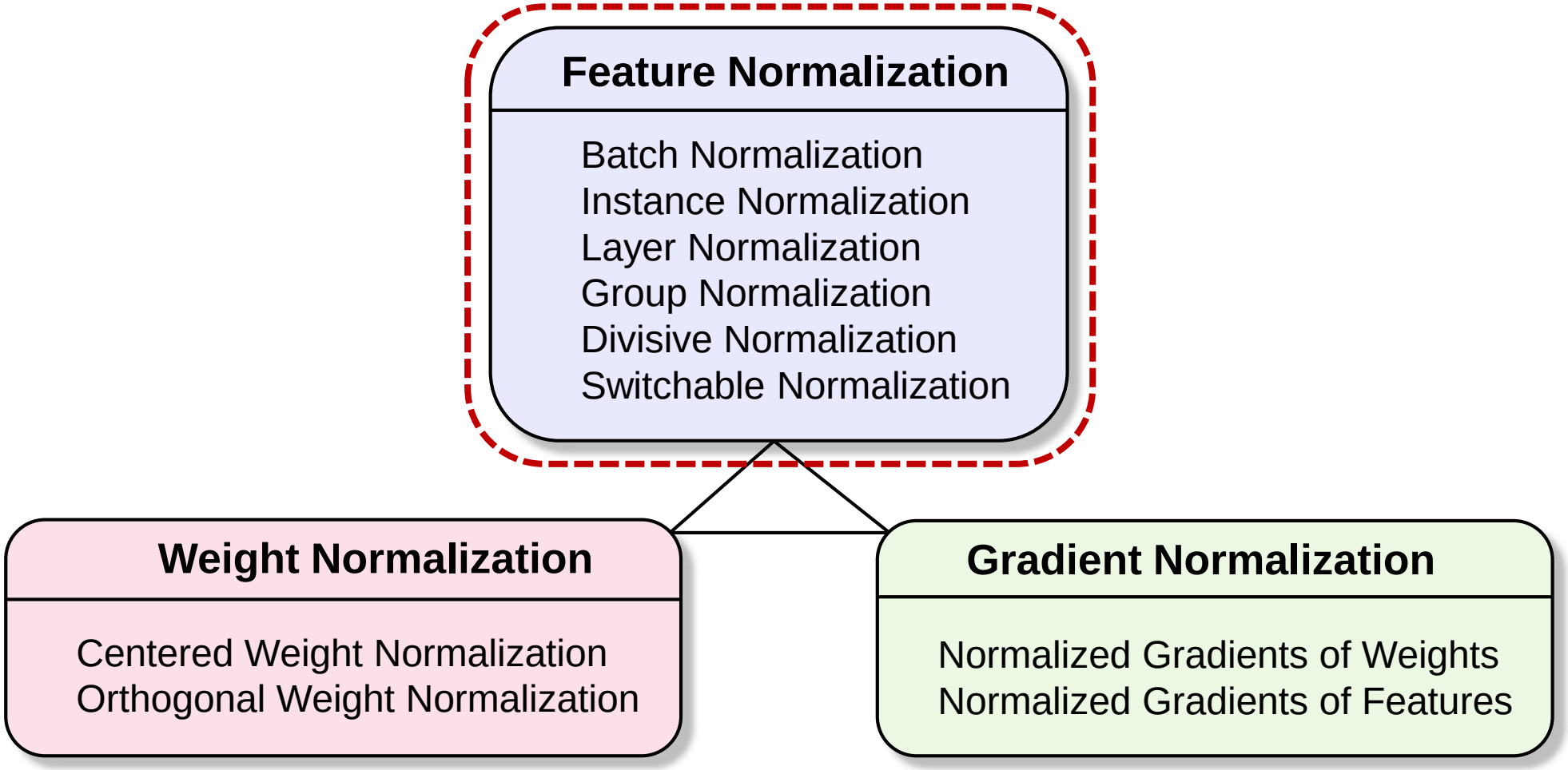
- **Combination of BN, IN and LN: Switchable Normalization**

$$\hat{x}_{nchw} = \gamma \frac{x_{nchw} - (w_{IN}\mu_{IN} + w_{BN}\mu_{BN} + w_{LN}\mu_{LN})}{\sqrt{w'_{IN}\sigma^2_{IN} + w'_{BN}\sigma^2_{BN} + w'_{LN}\sigma^2_{LN}}} + \beta$$

$$w_k = \frac{e^{\lambda_k}}{\sum_{z \in \{IN, LN, BN\}} e^{\lambda_z}}, k \in \{IN, LN, BN\}$$

# Taxonomy of Normalization Methods

## Normalization Methods in Deep Learning



# Weight Normalization

- **NIPS 2016 Paper by OpenAI**
  - **Title:** Weight Normalization: A Simple Reparameterization to Accelerate Training of Deep Neural Networks
  - **Main idea:**
    - **Normalize** the weight vectors in a model to **unit-norm** and introduce a **separate scalar** variable controlling the length
    - i.e., decouple the **direction** and **length** of weight vectors as **separate variables**

$$y = \phi(\mathbf{w} \cdot \mathbf{x} + b) \qquad \boxed{\mathbf{w} = \frac{g}{||\mathbf{v}||}\mathbf{v}}$$

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    - This provides the optimization algorithm with the flexibility to **rotate** the parameter vectors without affecting their respective **magnitudes** and **vice versa**.
      - Benefits:

        ① Faster convergence speed. ② Robust to learning rate. ③ Insensitive to noise.

# Gradient Normalization

- **ICML 2018 Paper**
  - **Title**: GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep **Multitask Networks**
  - **Main idea:**
    - Dynamically **tuning gradient magnitudes** to automatically **balance training** in multitask network
    - Penalize the network when backpropagated gradients from any task are **too large or too small**.

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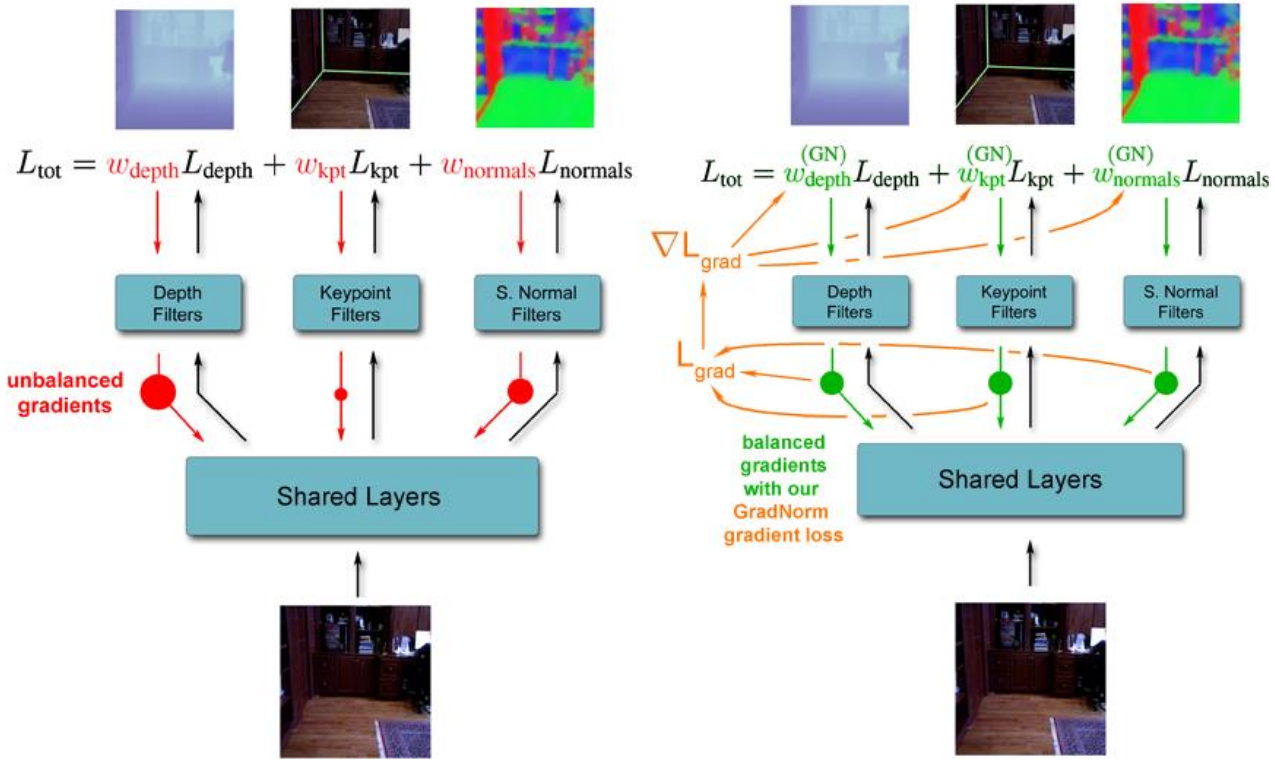

# Applying Normalization in Practice

- **Implementation**
  - Normalization is an independent module (view normalization as a **function**)
- **Empirical Tips**
  - For **Classification Model**:
    - Try **Batch Normalization** (it almost helps).
    - Try GN/LN/IN/DN if only **small batches** are required.
    - Try **weight normalization** if you expect to improve more.
  - For **Generative Model**:
    - Try **BN** and **IN**.

## Challenges

- **Black Art** of normalization methods:
  - **Dark Art** of **employment** of existing normalization methods.
    - Dependency on tasks, data, feature, network architecture and optimization methods
  - **Darker Art** of development of novel normalization methods.

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## Future Work

- To **employ** existing normalization methods efficiently.
- To **update** existing normalization methods appropriately.
- To **develop** novel normalization methods.

# Thanks!